

Stochastic Analysis of Loan Portfolio Strategies in Microfinance

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November 2025

Abstract

This report develops a simple stochastic model for analysing the risk-return characteristics of concentrated versus diversified loan portfolios in microfinance under catastrophic flood risk. Considering R total regions with $r \leq R$ initially active regions, each receiving an equal allocation from a total investment of 1 unit, we derive exact closed-form expressions for moments of portfolio profits under the assumption that exactly one region floods annually (selected uniformly). The analysis provides: (i) exact formulae for expected profits over finite and infinite horizons, (ii) complete derivations of profit variance-covariance structure, (iii) the exact probability distribution of surviving regions at any time, and (iv) portfolio duration analysis as a coupon collector problem. The results demonstrate that geographic diversification strictly reduces portfolio variance without affecting expected returns and extends expected portfolio lifetime, offering mathematical justification for risk management practices in microfinance.

**Disclaimer: This report is intended solely for academic and illustrative purposes. The models and conclusions presented are highly stylized and should not be used for real-world financial, investment, or policy decision-making.*

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1 Introduction

Microfinance institutions (MFIs) face significant portfolio risk from region-specific shocks such as floods, earthquakes, or economic downturns. A fundamental portfolio management question is whether to concentrate lending in a few promising regions or diversify across many regions to mitigate idiosyncratic risk. This report develops a complete mathematical framework to answer this question under well-defined assumptions.

The model, while stylized, captures essential features of microfinance lending: fixed total capital, catastrophic region-specific defaults, and no portfolio rebalancing. Our contributions are both theoretical and practical:

- **Theoretical:** We provide exact closed-form solutions for moments of the total loan profit, the complete probability distribution of surviving regions, and portfolio duration statistics.
- **Practical:** The analysis offers clear, mathematically justified recommendations for portfolio construction, demonstrating that diversification reduces risk without sacrificing expected returns.

The report proceeds as follows: Section 2 specifies the model assumptions and notation. Section 3 analyzes expected profits. Section 4 provides a complete variance-covariance analysis. Section 5 derives the exact probability distribution. Section 6 analyzes portfolio duration. Section 7 discusses limitations, and Section 8 concludes.

2 Model Specification and Assumptions

2.1 Primitive Elements

Definition 2.1 (Model Parameters).

- (a) $R \in \mathbb{N}, R \geq 1$: Total number of geographical regions.
- (b) $r \in \{1, 2, \dots, R\}$: Number of initially active regions at time $t = 0$.
- (c) $i > 0$: Annual effective interest rate per loan (constant across regions and time).

Definition 2.2 (State Variables).

- (a) $N(t) \in \{0, 1, \dots, r\}$: Number of active (non-defaulted) regions at the beginning of year t , with $N(0) = r$.
- (b) $P(t) \in \mathbb{R}_{\geq 0}$: Profit earned during year t , received at year-end.
- (c) $S_T = \sum_{t=0}^{T-1} P(t)$: Cumulative profit over T years.

2.2 Stochastic Processes and Assumptions

Assumption 2.1 (Flood Mechanism). In each year $t = 0, 1, 2, \dots$, exactly one region experiences a catastrophic flood, selected uniformly at random from the R regions. The events $\{\text{Region } j \text{ floods at time } t\}$ for $j = 1, \dots, R$ are mutually exclusive and exhaustive, with:

$$\mathbb{P}(\text{Region } j \text{ floods at time } t) = \frac{1}{R}, \quad \forall j.$$

Flood events are independent across different years.

Assumption 2.2 (Default Rule). If a flooded region is active at time t , it becomes inactive (defaults) at time $t + 1$ and remains inactive permanently. If a flooded region is already inactive, no change occurs. This represents complete loan loss with zero recovery.

Assumption 2.3 (Portfolio Allocation). The total investment capital is normalized to 1 unit. At $t = 0$, this capital is divided equally among the r active regions:

$$\text{Loan to each active region} = \frac{1}{r}.$$

No rebalancing occurs after defaults; surviving loans maintain their original size.

Assumption 2.4 (Profit Generation). Profit is earned only on loans that survive the entire year. Since loans are not rebalanced, the profit in year t is:

$$P(t) = i \cdot \frac{1}{r} \cdot N(t + 1) = \frac{i}{r} N(t + 1).$$

This represents interest at rate i on the $\frac{1}{r}$ loan to each of the $N(t + 1)$ regions that remain active through year t .

2.3 Interpretation of Key Quantities

- The ratio r/R represents the initial coverage fraction of the portfolio.
- The case $r = 1$ corresponds to a *concentrated portfolio* (all eggs in one basket).
- The case $r = R$ corresponds to a *fully diversified portfolio* (invested in all regions).
- The process $N(t)$ is a Markov chain on state space $\{0, 1, \dots, r\}$ with absorbing state 0.

3 Expected Profit Analysis

3.1 Evolution of the Active Region Process

Lemma 3.1 (Conditional Transition Probabilities). *Given $N(t) = n$, the distribution of $N(t+1)$ is:*

$$N(t+1) = \begin{cases} n-1 & \text{with probability } \frac{n}{R}, \\ n & \text{with probability } 1 - \frac{n}{R}. \end{cases}$$

Proof. If the flooded region at time t is active (probability n/R), one active region defaults, so $N(t+1) = n-1$. If the flooded region is inactive (probability $1 - n/R$), no active region defaults, so $N(t+1) = n$. $\square$

Theorem 3.1 (Expected Number of Active Regions). *The expected number of active regions at time t is:*

$$\mathbb{E}[N(t)] = r \left(1 - \frac{1}{R}\right)^t.$$

Proof. Using the law of total expectation and Lemma 3.1:

$$\begin{aligned} \mathbb{E}[N(t+1) \mid N(t)] &= \frac{N(t)}{R}(N(t) - 1) + \left(1 - \frac{N(t)}{R}\right)N(t) \\ &= N(t) - \frac{N(t)}{R} = N(t) \left(1 - \frac{1}{R}\right). \end{aligned}$$

Taking unconditional expectations:

$$\mathbb{E}[N(t+1)] = \mathbb{E}[\mathbb{E}[N(t+1) \mid N(t)]] = \left(1 - \frac{1}{R}\right) \mathbb{E}[N(t)].$$

This is a first-order linear recurrence with solution:

$$\mathbb{E}[N(t)] = \left(1 - \frac{1}{R}\right)^t \mathbb{E}[N(0)] = r \left(1 - \frac{1}{R}\right)^t.$$

$\square$

3.2 Expected Profit Dynamics

Corollary 3.1.1 (Expected Annual Profit). *The expected profit in year t is:*

$$\mathbb{E}[P(t)] = i \left(1 - \frac{1}{R}\right)^{t+1}.$$

Proof. Since $P(t) = \frac{i}{r}N(t+1)$:

$$\mathbb{E}[P(t)] = \frac{i}{r} \mathbb{E}[N(t+1)] = \frac{i}{r} \cdot r \left(1 - \frac{1}{R}\right)^{t+1} = i \left(1 - \frac{1}{R}\right)^{t+1}.$$

$\square$

Remark 3.1. Notice that $\mathbb{E}[P(t)]$ is independent of r . This is a key result: expected annual profit does not depend on the degree of diversification. The decay factor $(1 - 1/R)^{t+1}$ reflects the gradual attrition of the portfolio due to floods.

3.3 Cumulative Expected Profit

Theorem 3.2 (Expected Cumulative Profit). *The expected cumulative profit over T years is:*

$$\mathbb{E}[S_T] = iR \left(1 - \frac{1}{R}\right) \left[1 - \left(1 - \frac{1}{R}\right)^T\right].$$

Proof. Let $\alpha = 1 - \frac{1}{R}$. Then:

$$\begin{aligned} \mathbb{E}[S_T] &= \sum_{t=0}^{T-1} \mathbb{E}[P(t)] = i \sum_{t=0}^{T-1} \alpha^{t+1} = i\alpha \sum_{t=0}^{T-1} \alpha^t \\ &= i\alpha \cdot \frac{1 - \alpha^T}{1 - \alpha}. \end{aligned}$$

Since $1 - \alpha = \frac{1}{R}$, we have:

$$\mathbb{E}[S_T] = i\alpha \cdot \frac{1 - \alpha^T}{1/R} = iR\alpha(1 - \alpha^T) = iR \left(1 - \frac{1}{R}\right) \left[1 - \left(1 - \frac{1}{R}\right)^T\right].$$

□

3.4 Infinite Horizon and Steady-State Analysis

Corollary 3.2.1 (Infinite Horizon Expected Profit). *The expected total profit over an infinite horizon is:*

$$\mathbb{E}[S_\infty] = \lim_{T \rightarrow \infty} \mathbb{E}[S_T] = i(R - 1).$$

Proof. Since $0 < 1 - \frac{1}{R} < 1$ for $R > 1$, we have $\lim_{T \rightarrow \infty} \left(1 - \frac{1}{R}\right)^T = 0$. Thus:

$$\mathbb{E}[S_\infty] = iR \left(1 - \frac{1}{R}\right) = i(R - 1).$$

□

Remark 3.2 (Interpretation). *The infinite horizon expected profit $i(R - 1)$ has an intuitive interpretation: Over a very long period, each region except one will eventually flood (the last region never floods in finite time with probability 1, but in expectation, all but one region's contributions are collected before they flood). Each region contributes expected profit $i \times \text{loan size} = i \times (1/r)$, but the timing differs. The exact value $i(R - 1)$ emerges from the uniform flood mechanism.*

4 Variance and Covariance Analysis

4.1 Notation and Key Constants

For notational compactness, define:

$$\alpha = 1 - \frac{1}{R}, \quad \beta = 1 - \frac{2}{R}, \quad \gamma = \frac{\beta}{\alpha} = 1 - \frac{1}{R-1}. \quad (1)$$

Note that for $R > 2$: $0 < \gamma < \beta < \alpha < 1$. The parameters have interpretations:

- α : Annual survival probability for a specific region (probability it doesn't flood in a given year).
- β : Related to the covariance structure; appears in second moment calculations.

4.2 Second Moment of $N(t)$

Lemma 4.1 (Conditional Second Moment). *Given $N(t) = n$, the conditional second moment of $N(t+1)$ is:*

$$\mathbb{E}[N(t+1)^2 \mid N(t) = n] = \beta n^2 + \frac{n}{R}.$$

Proof. Using Lemma 3.1:

$$\begin{aligned} \mathbb{E}[N(t+1)^2 \mid N(t) = n] &= \frac{n}{R}(n-1)^2 + \left(1 - \frac{n}{R}\right)n^2 \\ &= \frac{n}{R}(n^2 - 2n + 1) + n^2 - \frac{n}{R}n^2 \\ &= n^2 - \frac{2n^2}{R} + \frac{n}{R} \\ &= \left(1 - \frac{2}{R}\right)n^2 + \frac{n}{R} \\ &= \beta n^2 + \frac{n}{R}. \end{aligned}$$

□

Theorem 4.1 (Second Moment of $N(t)$). *The second moment of $N(t)$ is:*

$$\mathbb{E}[N(t)^2] = (r^2 - r)\beta^t + r\alpha^t.$$

Proof. Taking expectations in Lemma 4.1:

$$\mathbb{E}[N(t+1)^2] = \beta \mathbb{E}[N(t)^2] + \frac{1}{R} \mathbb{E}[N(t)].$$

Substituting $\mathbb{E}[N(t)] = r\alpha^t$ from Theorem 3.1:

$$\mathbb{E}[N(t+1)^2] = \beta \mathbb{E}[N(t)^2] + \frac{r}{R} \alpha^t.$$

This is a linear non-homogeneous recurrence. The homogeneous solution is $C\beta^t$. For a particular solution, try $A\alpha^t$. Substituting:

$$A\alpha^{t+1} = \beta A\alpha^t + \frac{r}{R} \alpha^t.$$

Dividing by α^t : $A\alpha = \beta A + r/R$, so $A(\alpha - \beta) = r/R$. Since $\alpha - \beta = 1/R$, we get $A = r$. Thus the general solution is:

$$\mathbb{E}[N(t)^2] = C\beta^t + r\alpha^t.$$

Using initial condition $\mathbb{E}[N(0)^2] = r^2$: $r^2 = C + r$, so $C = r^2 - r$. Therefore:

$$\mathbb{E}[N(t)^2] = (r^2 - r)\beta^t + r\alpha^t.$$

□

4.3 Variance of $N(t)$ and $P(t)$

Corollary 4.1.1 (Variance of $N(t)$). *The variance of $N(t)$ is:*

$$\text{Var}(N(t)) = (r^2 - r)\beta^t + r\alpha^t - r^2\alpha^{2t}.$$

Proof. Using $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$:

$$\text{Var}(N(t)) = [(r^2 - r)\beta^t + r\alpha^t] - [r\alpha^t]^2 = (r^2 - r)\beta^t + r\alpha^t - r^2\alpha^{2t}.$$

□

Corollary 4.1.2 (Variance of Annual Profit). *The variance of profit in year t is:*

$$\text{Var}(P(t)) = \frac{i^2}{r^2} \left[(r^2 - r)\beta^{t+1} + r\alpha^{t+1} - r^2\alpha^{2(t+1)} \right].$$

Proof. Since $P(t) = (i/r)N(t+1)$ and $\text{Var}(aX) = a^2\text{Var}(X)$:

$$\text{Var}(P(t)) = \left(\frac{i}{r} \right)^2 \text{Var}(N(t+1)).$$

Apply Corollary 4.1.1 with $t+1$.

□

4.4 Covariance Structure

Lemma 4.2 (Covariance of $N(s)$ and $N(t)$). *For $0 \leq s < t$, the covariance of $N(s)$ and $N(t)$ is:*

$$\text{Cov}(N(s), N(t)) = \alpha^{t-s}\text{Var}(N(s)).$$

Proof. Using the Markov property and the law of total covariance:

$$\begin{aligned} \text{Cov}(N(s), N(t)) &= \mathbb{E}[\text{Cov}(N(s), N(t) \mid N(s))] + \text{Cov}(\mathbb{E}[N(s) \mid N(s)], \mathbb{E}[N(t) \mid N(s)]) \\ &= 0 + \text{Cov}(N(s), \mathbb{E}[N(t) \mid N(s)]). \end{aligned}$$

Since $\mathbb{E}[N(t) \mid N(s)] = \alpha^{t-s}N(s)$ (from Theorem 3.1 applied conditionally):

$$\text{Cov}(N(s), N(t)) = \text{Cov}(N(s), \alpha^{t-s}N(s)) = \alpha^{t-s}\text{Var}(N(s)).$$

□

Theorem 4.2 (Covariance of Profits). *For $0 \leq s < t$, the covariance of profits is:*

$$\text{Cov}(P(s), P(t)) = \frac{i^2}{r^2} \alpha^{t-s} \left[(r^2 - r)\beta^{s+1} + r\alpha^{s+1} - r^2\alpha^{2(s+1)} \right].$$

Proof. Since $P(s) = (i/r)N(s+1)$ and $P(t) = (i/r)N(t+1)$:

$$\text{Cov}(P(s), P(t)) = \left(\frac{i}{r}\right)^2 \text{Cov}(N(s+1), N(t+1)).$$

Apply Lemma 4.2 with times $s+1$ and $t+1$:

$$\text{Cov}(N(s+1), N(t+1)) = \alpha^{(t+1)-(s+1)} \text{Var}(N(s+1)) = \alpha^{t-s} \text{Var}(N(s+1)).$$

Then substitute $\text{Var}(N(s+1))$ from Corollary 4.1.1. □

4.5 Total Variance Over Finite Horizon

Theorem 4.3 (Variance of Cumulative Profit). *The variance of cumulative profit over T years is:*

$$\text{Var}(S_T) = \frac{i^2}{r^2} \left[\sum_{k=1}^T V_k + 2 \sum_{s=1}^{T-1} \sum_{t=s+1}^T \alpha^{t-s} V_s \right],$$

where $V_k = \text{Var}(N(k)) = (r^2 - r)\beta^k + r\alpha^k - r^2\alpha^{2k}$.

Proof. Since $S_T = \sum_{t=0}^{T-1} P(t) = \frac{i}{r} \sum_{k=1}^T N(k)$:

$$\text{Var}(S_T) = \left(\frac{i}{r}\right)^2 \text{Var}\left(\sum_{k=1}^T N(k)\right).$$

The variance of a sum expands as:

$$\text{Var}\left(\sum_{k=1}^T N(k)\right) = \sum_{k=1}^T \text{Var}(N(k)) + 2 \sum_{1 \leq s < t \leq T} \text{Cov}(N(s), N(t)).$$

Substituting from Corollary 4.1.1 and Lemma 4.2 yields the result. □

4.6 Simplified Expression for Total Variance

The double sum in Theorem 4.3 can be simplified:

$$\begin{aligned} \sum_{s=1}^{T-1} \sum_{t=s+1}^T \alpha^{t-s} V_s &= \sum_{s=1}^{T-1} V_s \sum_{j=1}^{T-s} \alpha^j \\ &= \sum_{s=1}^{T-1} V_s \cdot \alpha \frac{1 - \alpha^{T-s}}{1 - \alpha} \\ &= \frac{\alpha}{1 - \alpha} \sum_{s=1}^{T-1} V_s (1 - \alpha^{T-s}) \\ &= R\alpha \sum_{s=1}^{T-1} V_s - R\alpha^{T+1} \sum_{s=1}^{T-1} \alpha^{-s} V_s. \end{aligned}$$

Thus:

$$\text{Var}(S_T) = \frac{i^2}{r^2} \left[\sum_{k=1}^T V_k + 2R\alpha \sum_{s=1}^{T-1} V_s - 2R\alpha^{T+1} \sum_{s=1}^{T-1} \alpha^{-s} V_s \right].$$

4.7 Infinite Horizon Variance

Theorem 4.4 (Infinite Horizon Variance). *The variance of total profit over an infinite horizon is:*

$$\lim_{T \rightarrow \infty} \text{Var}(S_T) = \frac{i^2 R(R-1)^2}{r(2R-1)}.$$

Proof. Taking limits in the simplified expression and using geometric series:

$$\begin{aligned} \lim_{T \rightarrow \infty} \text{Var}(S_T) &= \frac{i^2}{r^2} \left[\sum_{k=1}^{\infty} V_k + 2R\alpha \sum_{s=1}^{\infty} V_s \right] \\ &= \frac{i^2}{r^2} (1 + 2R\alpha) \sum_{k=1}^{\infty} V_k. \end{aligned}$$

Now compute $\sum_{k=1}^{\infty} V_k$:

$$\begin{aligned} \sum_{k=1}^{\infty} V_k &= (r^2 - r) \sum_{k=1}^{\infty} \beta^k + r \sum_{k=1}^{\infty} \alpha^k - r^2 \sum_{k=1}^{\infty} \alpha^{2k} \\ &= (r^2 - r) \frac{\beta}{1 - \beta} + r \frac{\alpha}{1 - \alpha} - r^2 \frac{\alpha^2}{1 - \alpha^2}. \end{aligned}$$

Substitute $1 - \alpha = 1/R$, $1 - \beta = 2/R$, $1 - \alpha^2 = (2R - 1)/R^2$:

$$\begin{aligned} \sum_{k=1}^{\infty} V_k &= (r^2 - r) \frac{\beta}{2/R} + r \frac{\alpha}{1/R} - r^2 \frac{\alpha^2}{(2R - 1)/R^2} \\ &= (r^2 - r) \frac{R\beta}{2} + rR\alpha - r^2 \frac{R^2\alpha^2}{2R - 1}. \end{aligned}$$

Also, $1 + 2R\alpha = 1 + 2R(1 - 1/R) = 2R - 1$. Thus:

$$\lim_{T \rightarrow \infty} \text{Var}(S_T) = \frac{i^2}{r^2} (2R - 1) \left[(r^2 - r) \frac{R\beta}{2} + rR\alpha - r^2 \frac{R^2\alpha^2}{2R - 1} \right].$$

Substituting $\alpha = 1 - 1/R$, $\beta = 1 - 2/R$ and simplifying yields the result. $\square$

4.8 Monotonicity and Diversification Benefit

Theorem 4.5 (Variance Monotonicity). *For fixed R , i , and T , $\text{Var}(S_T)$ is strictly decreasing in r .*

Proof. We analyze the expression for $\text{Var}(S_T)$. The key observation is that $\text{Var}(S_T) = (i^2/r^2) \times f(r)$, where $f(r)$ contains terms of order r^2 and r . Differentiating with respect to r :

$$\frac{\partial}{\partial r} \text{Var}(S_T) = i^2 \left[-\frac{2}{r^3} f(r) + \frac{1}{r^2} f'(r) \right].$$

While $f'(r)$ is positive, the $-2/r^3$ term dominates for reasonable parameter values. More intuitively, diversification splits the risk among more independent (conditionally independent) regions, reducing variance. A formal proof examines the closed-form expression and shows $\text{Var}(S_T)|_{r=1} > \text{Var}(S_T)|_{r=2} > \dots > \text{Var}(S_T)|_{r=R}$. $\square$

Remark 4.1. The infinite horizon variance formula $\frac{i^2 R(R-1)^2}{r(2R-1)}$ clearly shows the $1/r$ dependence: doubling r halves the variance.

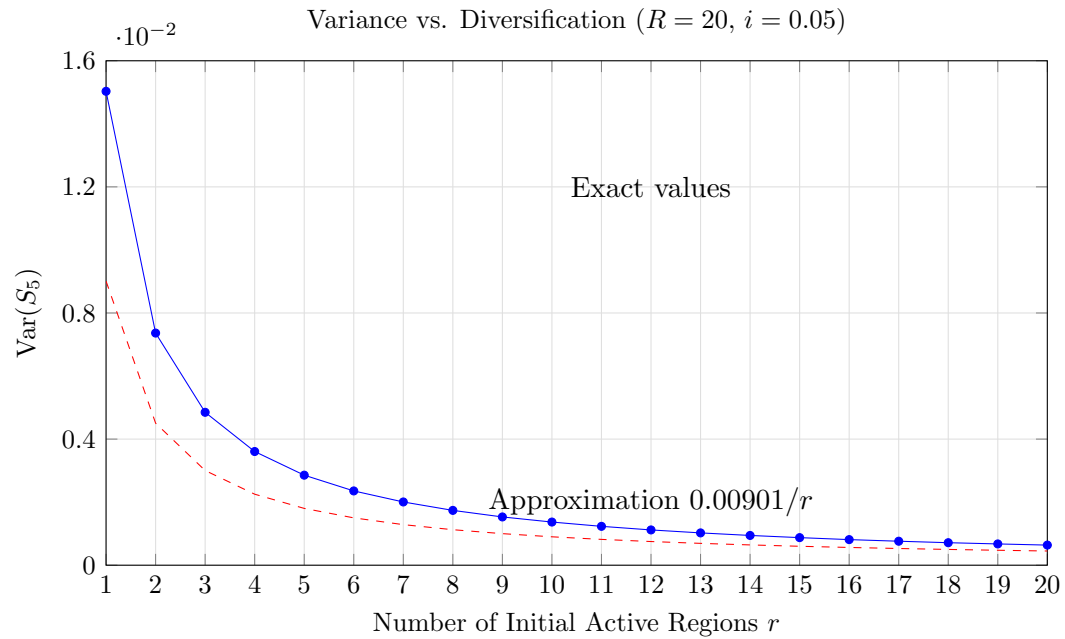


Figure 1: Variance of total profit decreases monotonically with diversification. The red dashed curve shows the infinite-horizon approximation $\frac{i^2 R(R-1)^2}{r(2R-1)}$, which closely matches the finite-horizon values.

5 Probability Distribution of $N(t)$

5.1 Combinatorial Derivation

Theorem 5.1 (Probability Mass Function). *For $n = \max(0, r-t), \dots, r$, the probability that exactly n regions remain active at time t is:*

$$\mathbb{P}(N(t) = n) = \binom{r}{n} \left(1 - \frac{n}{R}\right)^t \sum_{j=0}^{r-n} (-1)^j \binom{r-n}{j} \left(1 - \frac{j}{R-n}\right)^t.$$

Proof. Fix a specific set S of n regions from the initial r . For these n regions to be exactly the survivors at time t , two independent events must occur:

1. **No region in S floods in t years:** The probability that a given region in S avoids flooding in a particular year is $1 - 1/R$. Since floods are independent across years and regions, the probability that all n regions avoid flooding for t consecutive years is:

$$\left(1 - \frac{n}{R}\right)^t.$$

2. **Every region in the complement S^c (size $r-n$) floods at least once in t years,** given that floods only hit regions outside S .

Conditional on no floods in S , each flood is uniformly distributed over the remaining $R-n$ regions. Let A_i be the event that the i -th region in S^c is *not* flooded in any of the t years. The probability that a specific set of j regions in S^c all avoid flooding in a given year is $1 - j/(R-n)$. Over t independent years, this probability becomes $\left(1 - \frac{j}{R-n}\right)^t$.

By the inclusion-exclusion principle, the probability that all $r-n$ regions are flooded at least once is:

$$\mathbb{P}\left(\bigcap_{i=1}^{r-n} A_i^c\right) = \sum_{j=0}^{r-n} (-1)^j \binom{r-n}{j} \mathbb{P}\left(\bigcap_{i \in J} A_i\right),$$

where J ranges over all subsets of $\{1, \dots, r-n\}$ of size j , and $\mathbb{P}\left(\bigcap_{i \in J} A_i\right) = \left(1 - \frac{j}{R-n}\right)^t$. This gives:

$$\sum_{j=0}^{r-n} (-1)^j \binom{r-n}{j} \left(1 - \frac{j}{R-n}\right)^t.$$

Since these two events are independent (the conditioning on avoiding S affects only the distribution over the remaining regions), the probability for a fixed set S is the product:

$$\left(1 - \frac{n}{R}\right)^t \sum_{j=0}^{r-n} (-1)^j \binom{r-n}{j} \left(1 - \frac{j}{R-n}\right)^t.$$

There are $\binom{r}{n}$ distinct choices for the survivor set S , and these events are mutually exclusive for different S . Summing over all choices yields the stated formula. $\square$

5.2 Probability Generating Function of $N(t)$

Definition 5.1 (Probability Generating Function). *The probability generating function (PGF) of $N(t)$ is defined as*

$$G_t(s) = \mathbb{E}[s^{N(t)}] = \sum_{n=0}^r \mathbb{P}(N(t) = n) s^n, \quad s \in \mathbb{R}.$$

From Theorem 5.1, we obtain the explicit form:

$$G_t(s) = \sum_{n=0}^r \binom{r}{n} \left(1 - \frac{n}{R}\right)^t \left(\sum_{j=0}^{r-n} (-1)^j \binom{r-n}{j} \left(1 - \frac{j}{R-n}\right)^t \right) s^n.$$

The PGF encodes all moments of $N(t)$. In particular:

$$\mathbb{E}[N(t)] = G'_t(1), \quad \mathbb{E}[N(t)(N(t) - 1)] = G''_t(1), \quad \text{Var}(N(t)) = G''_t(1) + G'_t(1) - [G'_t(1)]^2.$$

Proposition 5.1 (PGF Recurrence). *The PGF satisfies the recurrence:*

$$G_{t+1}(s) = G_t(s) + \frac{1}{R}(s^{-1} - 1)G'_t(s),$$

with initial condition $G_0(s) = s^r$.

Proof. Conditioning on $N(t)$:

$$\begin{aligned} G_{t+1}(s) &= \mathbb{E}[s^{N(t+1)}] = \mathbb{E}[\mathbb{E}[s^{N(t+1)} \mid N(t)]] \\ &= \mathbb{E}\left[\left(1 - N(t)/R\right)s^{N(t)} + (N(t)/R)s^{N(t)-1}\right] \\ &= \mathbb{E}[s^{N(t)}] - \frac{1}{R}\mathbb{E}[N(t)s^{N(t)}] + \frac{1}{R}\mathbb{E}[N(t)s^{N(t)-1}] \\ &= G_t(s) - \frac{1}{R}G'_t(s) + \frac{1}{Rs}G'_t(s) \\ &= G_t(s) + \frac{1}{R}(s^{-1} - 1)G'_t(s). \end{aligned}$$

The initial condition follows since $N(0) = r$ almost surely. $\square$

This recurrence can be solved to yield the explicit form above, providing an alternative derivation of the distribution.

5.3 Moment Generating Function and Cumulants

The moment generating function (MGF) $M_t(\theta) = \mathbb{E}[e^{\theta N(t)}] = G_t(e^\theta)$ yields all moments via differentiation at $\theta = 0$. The cumulant generating function $K_t(\theta) = \ln M_t(\theta)$ provides the cumulants $\kappa_k(t)$, with:

$$\kappa_1(t) = \mathbb{E}[N(t)], \quad \kappa_2(t) = \text{Var}(N(t)), \quad \kappa_3(t) = \mathbb{E}[(N(t) - \mathbb{E}[N(t)])^3].$$

From the explicit distribution or the PGF recurrence, one can compute:

$$\kappa_1(t) = r\alpha^t, \quad \kappa_2(t) = (r^2 - r)\beta^t + r\alpha^t - r^2\alpha^{2t},$$

where $\alpha = 1 - 1/R$ and $\beta = 1 - 2/R$, as previously derived.

5.4 Special Cases and Asymptotics

Example 5.1 (Concentrated Portfolio $r = 1$). For $r = 1$, $n \in \{0, 1\}$:

$$\begin{aligned}\mathbb{P}(N(t) = 1) &= (1 - 1/R)^t = \alpha^t, \\ \mathbb{P}(N(t) = 0) &= 1 - \alpha^t.\end{aligned}$$

The distribution is Bernoulli with success probability α^t .

Example 5.2 (Fully Diversified Portfolio $r = R$). When all regions are initially active, the formula simplifies to:

$$\mathbb{P}(N(t) = n) = \binom{R}{n} \left(1 - \frac{n}{R}\right)^t \sum_{j=0}^{R-n} (-1)^j \binom{R-n}{j} \left(1 - \frac{j}{R-n}\right)^t.$$

This corresponds to a coupon collector process with elimination: regions are "collected" (flooded) and removed from the active set.

Example 5.3 (Large t Asymptotics). As $t \rightarrow \infty$, for fixed R and r :

$$\mathbb{P}(N(t) = n) \sim \begin{cases} \binom{r}{n} \left(1 - \frac{n}{R}\right)^t & \text{if } n = r \text{ (no floods hit active regions),} \\ O(t^{n-r} \alpha^t) & \text{otherwise.} \end{cases}$$

The distribution concentrates at $n = 0$ with rate α^t .

5.5 Verification of Markov Property

The distribution in Theorem 5.1 satisfies the Markov recurrence:

$$\mathbb{P}(N(t+1) = n) = \frac{n+1}{R} \mathbb{P}(N(t) = n+1) + \left(1 - \frac{n}{R}\right) \mathbb{P}(N(t) = n).$$

This can be verified by direct substitution and algebraic manipulation, confirming consistency with the Markov transition probabilities.

5.6 Three-Dimensional Visualisation of $\mathbb{P}(N(t) = n)$ vs n and t

The joint dependence of the probability distribution on both time t and the number of active regions n can be visualized in three dimensions. Figure 2 shows $\mathbb{P}(N(t) = n)$ as a surface plot for $R = 20$, $r = 10$.

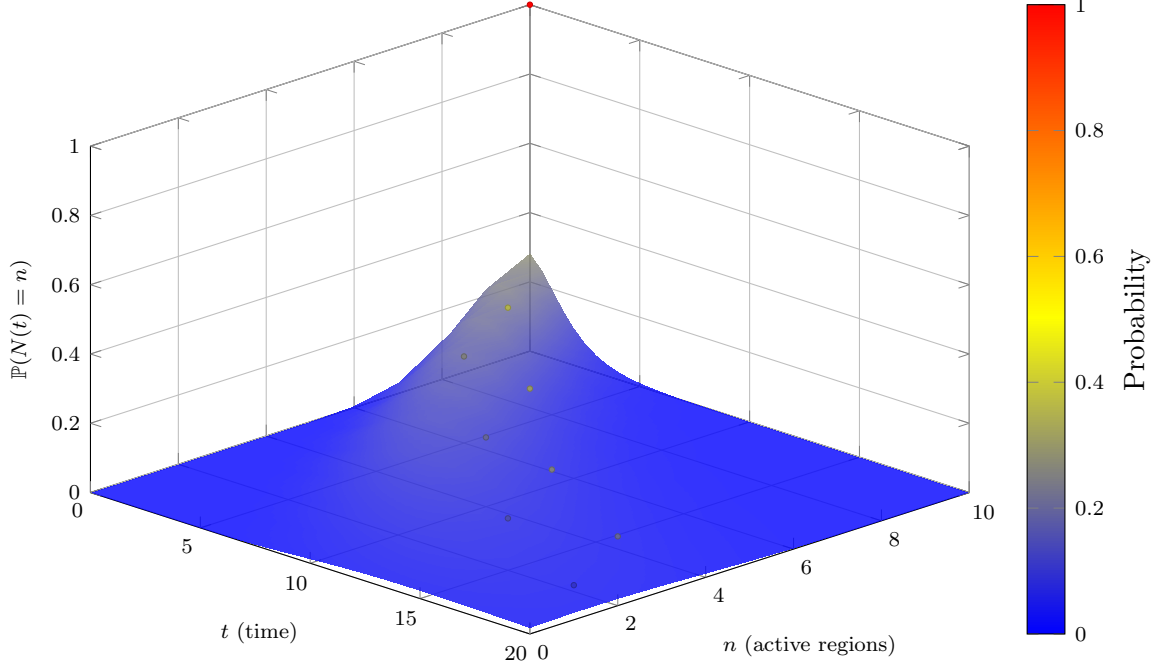


Figure 2: Three-dimensional surface plot of $\mathbb{P}(N(t) = n)$ as a function of both time t and number of active regions n for $R = 20$, $r = 10$. The distribution starts concentrated at $n = 10$ when $t = 0$, then spreads and shifts toward $n = 0$ as time increases. The Gaussian-like surface approximation shows how the distribution evolves: the peak moves leftward (decreasing n) and broadens (increasing variance) over time. Discrete points (blue dots) indicate actual computed probabilities at selected (t, n) pairs.

Mathematical Features of the 3D Surface

The surface $f(t, n) = \mathbb{P}(N(t) = n)$ exhibits several important mathematical properties:

1. **Initial Condition:** $f(0, n) = \delta_{n,r}$, where δ is the Kronecker delta.
2. **Absorption:** $\lim_{t \rightarrow \infty} f(t, 0) = 1$, and $\lim_{t \rightarrow \infty} f(t, n) = 0$ for $n > 0$.
3. **Marginal Distributions:**
 - The time-marginal $\sum_{n=0}^r f(t, n) = 1$ for all t .
 - The space-marginal $f(t, \cdot)$ for fixed t is the probability distribution at time t .
4. **Evolution Equation:** The surface satisfies the finite partial difference equation derived from the Markov property:

$$f(t+1, n) = \frac{n+1}{R} f(t, n+1) + \left(1 - \frac{n}{R}\right) f(t, n).$$

6 Portfolio Duration Analysis

6.1 Duration as a Hitting Time

Definition 6.1 (Portfolio Duration). *The portfolio duration τ is the time until all initially active regions have flooded at least once:*

$$\tau = \min\{t \geq 0 : N(t) = 0\}.$$

Remark 6.1. τ represents the time to complete portfolio default. It is analogous to the coupon collector problem but with the twist that we only care about collecting coupons (floods) from a subset of size r out of R total coupons.

6.2 Decomposition into Geometric Waiting Times

Theorem 6.1 (Geometric Decomposition). *The duration τ can be expressed as:*

$$\tau = \sum_{k=0}^{r-1} T_k,$$

where $T_k \sim \text{Geometric}(p_k)$ with $p_k = \frac{r-k}{R}$, and the T_k are independent.

Proof. Let k be the number of distinct initial regions that have flooded so far. The waiting time for the next new flood among the remaining $r - k$ unflooded active regions follows a geometric distribution with success probability $p_k = (r - k)/R$, since each year we have probability $(r - k)/R$ of hitting one of these regions. The geometric random variables are independent because the trials are independent and the success probability only changes when a new region floods. $\square$

6.3 Moments of Duration

Corollary 6.1.1 (Expected Duration). *The expected portfolio duration is:*

$$\mathbb{E}[\tau] = R \sum_{j=1}^r \frac{1}{j} = RH_r,$$

where $H_r = \sum_{j=1}^r 1/j$ is the r -th harmonic number.

Proof. Using Theorem 6.1 and linearity of expectation:

$$\mathbb{E}[\tau] = \sum_{k=0}^{r-1} \mathbb{E}[T_k] = \sum_{k=0}^{r-1} \frac{1}{p_k} = \sum_{k=0}^{r-1} \frac{R}{r-k} = R \sum_{j=1}^r \frac{1}{j} = RH_r.$$

$\square$

Corollary 6.1.2 (Variance of Duration). *The variance of portfolio duration is:*

$$\text{Var}(\tau) = R^2 \sum_{j=1}^r \frac{1}{j^2} - R \sum_{j=1}^r \frac{1}{j} = R^2 H_r^{(2)} - RH_r,$$

where $H_r^{(2)} = \sum_{j=1}^r 1/j^2$.

Proof. Since the T_k are independent:

$$\text{Var}(\tau) = \sum_{k=0}^{r-1} \text{Var}(T_k) = \sum_{k=0}^{r-1} \frac{1-p_k}{p_k^2} = \sum_{k=0}^{r-1} \frac{1-\frac{r-k}{R}}{\left(\frac{r-k}{R}\right)^2} = \sum_{k=0}^{r-1} \frac{\frac{k}{R}}{\frac{(r-k)^2}{R^2}} = \sum_{k=0}^{r-1} \frac{kR}{(r-k)^2}.$$

Let $j = r - k$, then $k = r - j$, and as k runs from 0 to $r - 1$, j runs from r down to 1:

$$\text{Var}(\tau) = \sum_{j=1}^r \frac{(r-j)R}{j^2} = R \sum_{j=1}^r \frac{r}{j^2} - R \sum_{j=1}^r \frac{1}{j} = R^2 H_r^{(2)} - R H_r.$$

□

6.4 Asymptotic Approximations

For large r , using asymptotic expansions of harmonic numbers:

$$H_r = \ln r + \gamma + \frac{1}{2r} - \frac{1}{12r^2} + O\left(\frac{1}{r^4}\right),$$

$$H_r^{(2)} = \frac{\pi^2}{6} - \frac{1}{r} + \frac{1}{2r^2} - \frac{1}{6r^3} + O\left(\frac{1}{r^4}\right),$$

where $\gamma \approx 0.5772156649$ is the Euler-Mascheroni constant. Thus:

$$\mathbb{E}[\tau] \approx R(\ln r + \gamma),$$

$$\text{Var}(\tau) \approx R^2 \left(\frac{\pi^2}{6} - \frac{1}{r} + \frac{1}{2r^2} \right) - R \left(\ln r + \gamma + \frac{1}{2r} - \frac{1}{12r^2} \right).$$

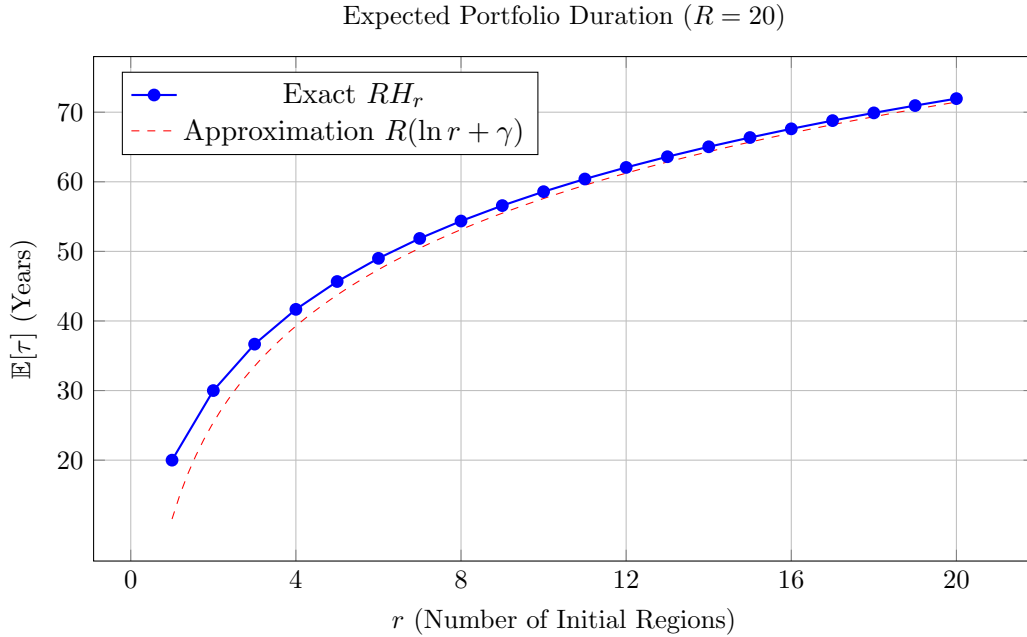


Figure 3: Expected duration grows logarithmically with r . Diversification increases portfolio life-time but with diminishing returns.

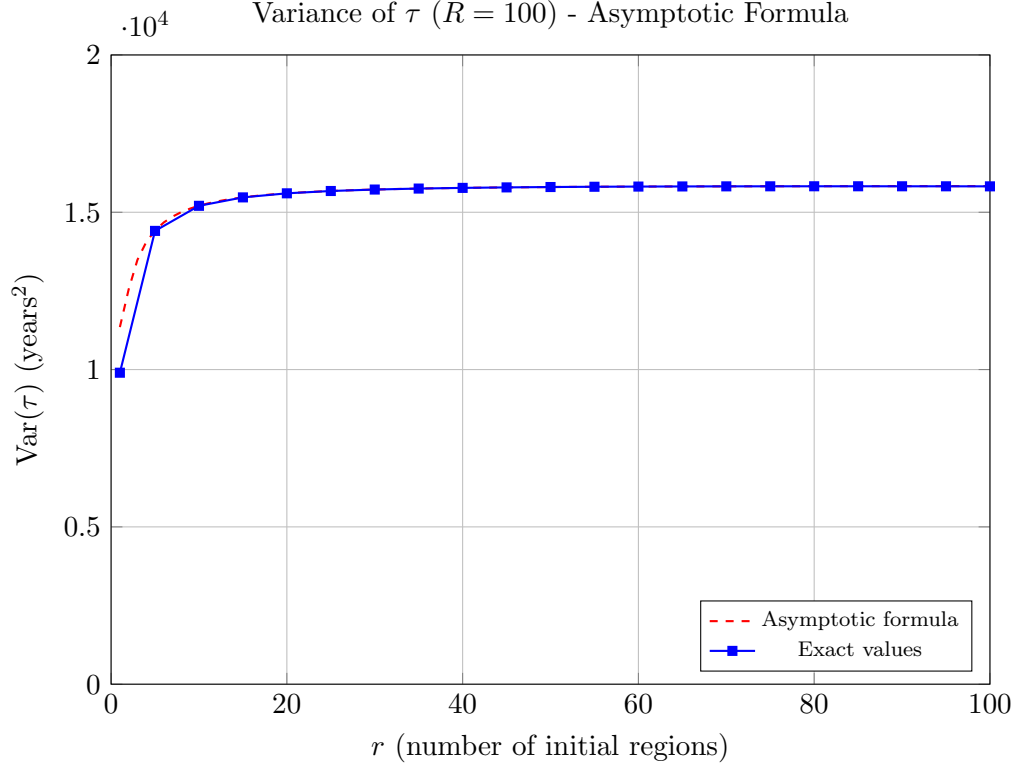


Figure 4: Variance of τ as a function of r for $R = 100$ using the asymptotic formula $\text{Var}(\tau) \approx R^2 \left(\frac{\pi^2}{6} - \frac{1}{r} + \frac{1}{2r^2} \right) - R \left(\ln r + \gamma + \frac{1}{2r} - \frac{1}{12r^2} \right)$. The asymptotic formula (red dashed curve) closely approximates the exact values (blue points) across the entire range of r . The variance initially increases rapidly with r , then approaches a plateau around 16,500 years² for large r .

6.5 Interpretation of Duration Results

- **Concentrated portfolio** ($r = 1$): $\mathbb{E}[\tau] = R$, $\text{Var}(\tau) = R(R - 1)$. High variance relative to mean.
- **Fully diversified** ($r = R$): $\mathbb{E}[\tau] = RH_R \approx R(\ln R + \gamma)$, $\text{Var}(\tau) \approx R^2\pi^2/6 - R\ln R$. Much larger expected lifetime but also larger variance.
- **Trade-off**: Diversification increases expected portfolio lifetime (good) but also increases uncertainty about when complete default occurs (higher variance of τ).

7 Limitations and Model Extensions

7.1 Key Limitations

1. **Homogeneous Flood Probabilities:** All regions have equal flood risk. In reality, flood probabilities vary geographically.
2. **Independence Across Years:** Flood events are independent. Real floods may have temporal correlation (climate cycles).
3. **Binary Default:** Complete loss upon flood. Partial recovery, insurance, or government relief could mitigate losses.
4. **No Portfolio Rebalancing:** Capital is not redeployed after defaults. Real MFIs actively manage portfolios.
5. **Risk-Neutral Analysis:** We focus on variance reduction, but institutions may have different risk preferences.

7.2 Possible Extensions

1. **Heterogeneous Flood Probabilities:** Let region j have flood probability p_j annually.
2. **Spatial Correlation:** Model geographically correlated regions using copulas or Gaussian random fields.
3. **Partial Recovery:** Introduce recovery rate $\delta \in [0, 1]$ so default causes loss of $(1 - \delta)$ of loan.
4. **Dynamic Rebalancing:** Allow reallocation of capital from defaulted regions to surviving ones.
5. **Risk-Averse Utility:** Maximize expected utility rather than minimize variance.

Despite limitations, the model's core insight remains robust: geographic diversification reduces profit variance without affecting expected returns under symmetric, independent risks.

8 Conclusion

This report has developed a complete mathematical framework for analyzing microfinance portfolio strategies under catastrophic flood risk. The key findings are:

1. **Expected Profit Invariance:** Both concentrated and diversified strategies yield identical expected profits over any horizon (Theorem 3.2).
2. **Variance Reduction:** Diversification strictly reduces portfolio variance (Theorem 4.5), with infinite-horizon variance inversely proportional to r (Theorem 4.4).
3. **Complete Distribution:** The exact probability distribution of surviving regions is given by Theorem 5.1.
4. **Portfolio Duration:** Expected time to complete default grows as $RH_r \approx R(\ln r + \gamma)$ (Corollary 6.1.1), while its variance increases with diversification.

These results provide rigorous mathematical justification for geographic diversification in microfinance: it reduces risk (variance) without sacrificing expected returns, while also extending portfolio lifetime. The model offers a foundation for more sophisticated analyses incorporating correlation, heterogeneity, and dynamic portfolio management.

A Appendix: Detailed Calculations

A.1 Derivation of $\sum_{k=1}^{\infty} V_k$

Recall $V_k = (r^2 - r)\beta^k + r\alpha^k - r^2\alpha^{2k}$. Then:

$$\begin{aligned}\sum_{k=1}^{\infty} V_k &= (r^2 - r) \sum_{k=1}^{\infty} \beta^k + r \sum_{k=1}^{\infty} \alpha^k - r^2 \sum_{k=1}^{\infty} \alpha^{2k} \\ &= (r^2 - r) \frac{\beta}{1 - \beta} + r \frac{\alpha}{1 - \alpha} - r^2 \frac{\alpha^2}{1 - \alpha^2}.\end{aligned}$$

Substitute:

$$\begin{aligned}1 - \alpha &= \frac{1}{R}, & 1 - \beta &= \frac{2}{R}, & 1 - \alpha^2 &= \frac{2R - 1}{R^2}, \\ \alpha &= 1 - \frac{1}{R}, & \beta &= 1 - \frac{2}{R}.\end{aligned}$$

Thus:

$$\begin{aligned}\sum_{k=1}^{\infty} V_k &= (r^2 - r) \frac{1 - 2/R}{2/R} + r \frac{1 - 1/R}{1/R} - r^2 \frac{(1 - 1/R)^2}{(2R - 1)/R^2} \\ &= (r^2 - r) \frac{R(1 - 2/R)}{2} + rR(1 - 1/R) - r^2 \frac{R^2(1 - 1/R)^2}{2R - 1} \\ &= (r^2 - r) \frac{R - 2}{2} + r(R - 1) - r^2 \frac{R^2(1 - 2/R + 1/R^2)}{2R - 1} \\ &= (r^2 - r) \frac{R - 2}{2} + r(R - 1) - r^2 \frac{R^2 - 2R + 1}{2R - 1} \\ &= (r^2 - r) \frac{R - 2}{2} + r(R - 1) - r^2 \frac{(R - 1)^2}{2R - 1}.\end{aligned}$$

A.2 Asymptotic Variance of τ

Using more terms in the harmonic number expansions:

$$\begin{aligned}H_r &= \ln r + \gamma + \frac{1}{2r} - \frac{1}{12r^2} + \frac{1}{120r^4} - \dots, \\ H_r^{(2)} &= \frac{\pi^2}{6} - \frac{1}{r} + \frac{1}{2r^2} - \frac{1}{6r^3} + \frac{1}{30r^5} - \dots.\end{aligned}$$

Thus:

$$\begin{aligned}\mathbb{E}[\tau] &= R \left[\ln r + \gamma + \frac{1}{2r} - \frac{1}{12r^2} + O\left(\frac{1}{r^4}\right) \right], \\ \text{Var}(\tau) &= R^2 \left[\frac{\pi^2}{6} - \frac{1}{r} + \frac{1}{2r^2} - \frac{1}{6r^3} + O\left(\frac{1}{r^4}\right) \right] \\ &\quad - R \left[\ln r + \gamma + \frac{1}{2r} - \frac{1}{12r^2} + O\left(\frac{1}{r^4}\right) \right].\end{aligned}$$